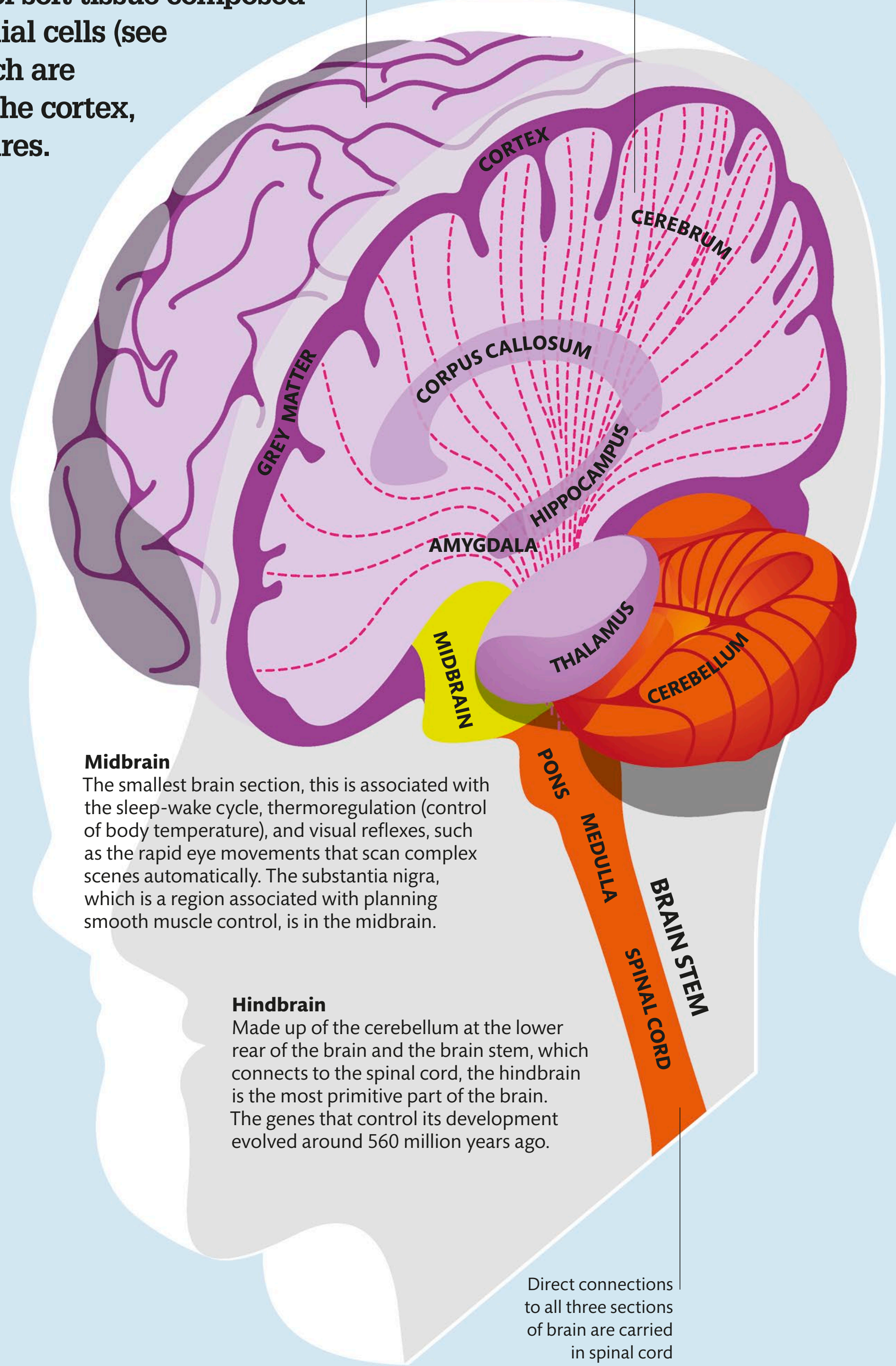
Brain Anatomy

The brain is a complex mass of soft tissue composed almost entirely of neurons, glial cells (see p.21), and blood vessels, which are grouped into an outer layer, the cortex, and other specialized structures.

Divisions of the brain

The brain is divided into three unequal parts: the forebrain, midbrain, and hindbrain. These divisions are based on how they develop in the embryonic brain, but they also reflect differences in function. In the human brain, the forebrain dominates, making up nearly 90 percent of the brain by weight. It is associated with sensory perception and higher executive functions. The midbrain and hindbrain below it are more involved with the basic bodily functions that determine survival, such as sleep and alertness.



Midbrain
The smallest brain section, this is associated with the sleep-wake cycle, thermoregulation (control of body temperature), and visual reflexes, such as the rapid eye movements that scan complex scenes automatically. The substantia nigra, which is a region associated with planning smooth muscle control, is in the midbrain.

Hindbrain
Made up of the cerebellum at the lower rear of the brain and the brain stem, which connects to the spinal cord, the hindbrain is the most primitive part of the brain. The genes that control its development evolved around 560 million years ago.

SPINAL NERVES

There are 31 pairs of spinal nerves that branch out from the spinal cord above each vertebral bone, named after the parts of the spine to which they connect. They relay signals between the brain and sensory organs, muscles, and glands.